Closing the Foveal Gap: Perceptually Grounded Scanpath Comparison with Disc IoU

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Figure 1: (a) Traditional reductive representation of scanpath fixations, (b) FDISS Foveal disc representation of fixations.

ABSTRACT

Scanpath comparison metrics universally represent fixations as dimensionless (x, y) points, despite the human fovea subtending approximately $1^\circ \sim 2^\circ$ of visual angle. This mismatch penalises spatially proximate fixations that process identical visual content. We propose **FDISS** (Foveal Disc IoU Scanpath Score), a metric that models each fixation as a foveal disc and computes similarity via bidirectional nearest-neighbour IoU matching, yielding interpretable Precision, Recall, and F1 scores. FDISS introduces a biologically grounded perceptual tolerance zone absent from all existing metrics and requires no free parameters at runtime.[1]

CCS CONCEPTS

• **Human-centered computing → Empirical studies in HCI**.

KEYWORDS

scanpath similarity; fixation metrics; foveal disc; eye tracking; IoU

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[1]https://github.com/kmamine/FDISS

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1 INTRODUCTION

Every dominant scanpath metric(DTW, MultiMatch [Jarodzka et al. 2010], ScanMatch [Cristino et al. 2010], ... etc.) treats a fixation as a point. Similarity reduces to Euclidean distance between coordinate pairs. The implicit assumption is that gaze precision is pixel-perfect, yet the human fovea subtends ~ 1–2° of visual angle [Bruno et al. 2023; Kerkouri et al. 2024; Tliba et al. 2022a; Wong et al. 2025]. Two observers fixating the same word 15 pixels apart are attending to identical content; a point metric penalises them identically to observers 200 pixels apart.

The community is aware of foveal extent: Gaussian kernels of $\sigma = 1^\circ$ are standard when constructing saliency maps [Bylinskii et al. 2019]. Yet this knowledge has never migrated into scanpath *comparison*. The result is a systematic inflation of dissimilarity for spatially proximate fixations, a limitation that directly affects the evaluation of learned scanpath prediction models [Kerkouri et al. 2022a, 2026, 2021, 2022b; Tliba et al. 2022a,b], where spatially proximate predictions are incorrectly penalised despite attending to perceptually identical content.

We close this gap with **FDISS**, a scanpath similarity metric that replaces the point primitive with a foveal disc and borrows the intersection-over-union (IoU) matching logic from object detection evaluation (mAP, PASCAL VOC [Everingham et al. 2010]) and multi-object tracking (MOTA/MOTP [Bernardin and Stiefelhagen 2008]).

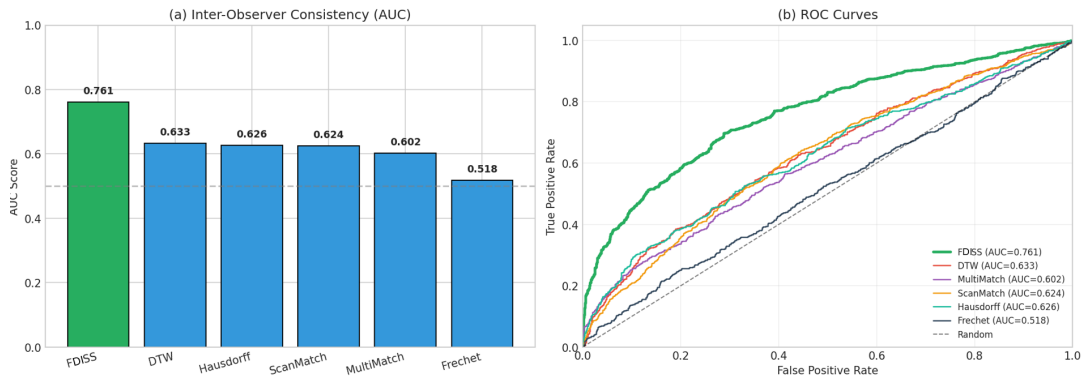

Figure 2: (a) Inter-observer consistency AUC comparing metrics. (b) ROC Curve of the metrics.

2 METHOD

2.1 Fixation as a Foveal Disc

Standard metrics represent each fixation $f_i = (x_i, y_i, d_i)$ as a point in image space. We instead represent it as a *foveal disc*:

$$\mathrm{Disc}(f_i) = \left\{(x, y) \in \mathbb{R}^2 \mid (x - x_i)^2 + (y - y_i)^2 \le r^2\right\}$$

where r is the foveal radius in pixels, computed once per dataset from its viewing geometry:

$$r = \tan\left(\tfrac{\pi}{180}\right) \cdot D_{\mathrm{ppcm}} \cdot D_{\mathrm{cm}}$$

with D_{ppcm} the screen resolution (px/cm) and D_{cm} the viewing distance. r is constant across all fixations in a dataset; FDISS has **no free parameters** at runtime.

2.2 Disc IoU

Let $d_{ij} = \|(x_i - x_j,\ y_i - y_j)\|_2$ be the Euclidean distance between two fixation centres. The intersection area of two equal-radius discs has a closed form:

$$\mathcal{I}(d_{ij}) = \begin{cases} \pi r^2 & d_{ij} = 0 \\ 2r^2 \arccos\left(\dfrac{d_{ij}}{2r}\right) - \dfrac{d_{ij}}{2}\sqrt{4r^2 - d_{ij}^2} & 0 < d_{ij} < 2r \\ 0 & d_{ij} \ge 2r \end{cases}$$

The **Disc IoU** between fixations f_i and g_j is then:

$$\mathrm{IoU}(f_i, g_j) = \frac{\mathcal{I}(d_{ij})}{2\pi r^2 - \mathcal{I}(d_{ij})} \in [0, 1]$$

No numerical integration is required. A key consequence of this formulation is a **hard perceptual boundary**: IoU = 0 whenever $d_{ij} \ge 2r$ ($\approx 2^\circ$), so fixations more than 2° apart are never matched regardless of the alignment strategy.

2.3 Bidirectional Nearest-Neighbour Matching

Given $S_A = \{f_1, \ldots, f_n\}$ and $S_B = \{g_1, \ldots, g_m\}$, the two comparing fixation sequences **A** and **B** we build the IoU matrix:

$$M \in [0, 1]^{n \times m}, \qquad M_{ij} = \mathrm{IoU}(f_i, g_j)$$

and match each fixation to its *best-overlapping* counterpart in the other scanpath. We adopt **many-to-one** nearest-neighbour rather than one-to-one (Hungarian) assignment: if two fixations in S_B both overlap a fixation in S_A, both receive full credit, reflecting attentional dwell rather than discrete object-level matching.

2.4 Precision, Recall, and FDISS

Matching is performed in both directions independently:

$$P(S_A, S_B) = \frac{1}{n}\sum_{i=1}^{n} \max_j M_{ij} \qquad R(S_A, S_B) = \frac{1}{m}\sum_{j=1}^{m} \max_i M_{ij}$$

Precision asks: *"for every fixation in S_A, did S_B look there?"* Recall asks the converse: *"for every fixation in S_B, did S_A look there?"* Note that $P(S_A, S_B) = R(S_B, S_A)$, since IoU depends only on d_{ij}; this ensures FDISS is symmetric without additional constraints. The final score is their harmonic mean:

$$\mathrm{FDISS}(S_A, S_B) = \frac{2 \cdot P \cdot R}{P + R}$$

The harmonic mean penalises imbalance between the two directions, collapsing toward zero whenever either P or R is low regardless of the other. FDISS is symmetric, bounded in $[0, 1]$, equals 1 if the scanpaths are identical, equals 0 if all pairwise disc distances exceed $2r$, and runs in $O(nm)$ per pair.

3 EXPERIMENTS

3.1 Setup

We evaluate on 100 randomly sampled images from CocoFreeView [Chen et al. 2022; Yang et al. 2023], each accompanied by 5 recorded scanpaths, yielding 1,000 same-image and 1,000 different-image pairs. The foveal radius is set to $r = 45$ px, corresponding to 1° of visual angle at the dataset's native viewing geometry (45 px/degree). No further tuning is performed. As our evaluation criterion we adopt **inter-observer consistency AUC**: the area under the ROC curve obtained by using each metric to separate same-image from different-image pairs, where a higher AUC indicates that the metric better reflects shared attentional structure across observers.

3.2 Results

FDISS achieves an AUC of 0.761, outperforming all baselines by 12.8–24.3 points and running 6–21× faster than DTW (Table 1). The advantage stems directly from the foveal disc primitive: point-based

Table 1: Inter-observer consistency AUC and per-pair runtime on CocoFreeView. All FDISS vs. baseline differences are significant at $p < 0.001$ (Bonferroni-corrected pairwise Mann–Whitney U tests).

Metric	AUC	95% CI
FDISS (ours)	**0.761**	[0.740, 0.780]
DTW	0.633	[0.609, 0.658]
ScanMatch	0.625	[0.600, 0.648]
Hausdorff	0.626	[0.601, 0.649]
MultiMatch	0.602	[0.578, 0.626]
Fréchet	0.518	[0.493, 0.544]

metrics penalise spatially proximate fixations that share perceptual content, inflating dissimilarity and degrading discriminability.

The hard perceptual boundary is empirically confirmed: Disc IoU falls continuously from 0.508 at $d = 21$ px to exactly zero at $d \approx 90$ px ($= 2r$), with no matched pairs beyond that threshold. A sensitivity analysis over $r \in [30, 90]$ px yields AUC in the range $[0.726, 0.801]$, confirming that FDISS is robust to reasonable variation in viewing-geometry assumptions and does not require precise calibration of the foveal radius.

4 CONCLUSION

FDISS replaces the point-fixation primitive with a biologically grounded foveal disc, enabling scanpath comparison that respects the perceptual resolution of the visual system. The resulting metric is parameter-free, analytically tractable, and produces interpretable Precision/Recall components. We release the implementation openly to facilitate adoption alongside existing methods.

REFERENCES

Keni Bernardin and Rainer Stiefelhagen. 2008. Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics. *EURASIP Journal on Image and Video Processing* 2008 (2008), 1–10.

Alessandro Bruno, Marouane Tliba, Mohamed Amine Kerkouri, Aladine Chetouani, Carlo Calogero Giunta, and Arzu Çöltekin. 2023. Detecting colour vision deficiencies via Webcam-based Eye-tracking: A case study. In *Proceedings of the 2023 Symposium on Eye Tracking Research and Applications* (Tubingen, Germany) *(ETRA '23)*. Association for Computing Machinery, New York, NY, USA, Article 39, 2 pages. https://doi.org/10.1145/3588015.3590133

Zoya Bylinskii, Tilke Judd, Aude Oliva, Antonio Torralba, and Frédo Durand. 2019. What Do Different Evaluation Metrics Tell Us About Saliency Models? *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 3 (2019), 740–757.

Yupei Chen, Zhibo Yang, Souradeep Chakraborty, Sounak Mondal, Seoyoung Ahn, Dimitris Samaras, Minh Hoai, and Gregory Zelinsky. 2022. Characterizing Target-Absent Human Attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 5031–5040.

Filipe Cristino, Sebastiaan Mathôt, Jan Theeuwes, and Iain D. Gilchrist. 2010. ScanMatch: A novel method for comparing fixation sequences. *Behavior Research Methods* 42, 3 (2010), 692–700.

Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. 2010. The Pascal Visual Object Classes (VOC) Challenge. *International Journal of Computer Vision* 88, 2 (2010), 303–338.

Halszka Jarodzka, Kenneth Holmqvist, and Marcus Nyström. 2010. A vector-based, multidimensional scanpath similarity measure. In *Proceedings of the 2010 Symposium on Eye-Tracking Research & Applications (ETRA '10)*. ACM, 211–218.

Mohamed Amine Kerkouri, Marouane Tliba, Aladine Chetouani, and Alessandro Bruno. 2022a. A domain adaptive deep learning solution for scanpath prediction of paintings. In *Proceedings of the 19th International Conference on Content-Based Multimedia Indexing*. 57–63.

Mohamed Amine Kerkouri, Marouane Tliba, Aladine Chetouani, and Alessandro Bruno. 2024. AVAtt: Art Visual Attention dataset for diverse painting styles. In *Proceedings of the 2024 Symposium on Eye Tracking Research and Applications* (Glasgow, United Kingdom) *(ETRA '24)*. Association for Computing Machinery, New York, NY, USA, Article 34, 3 pages. https://doi.org/10.1145/3649902.3655656

Mohamed Amine Kerkouri, Marouane Tliba, Aladine Chetouani, and Alessandro Bruno. 2026. SPGen: Stochastic scanpath generation for paintings using unsupervised domain adaptation. *arXiv preprint arXiv:2602.22049* (2026).

Mohamed A Kerkouri, Marouane Tliba, Aladine Chetouani, and Rachid Harba. 2021. Salypath: A deep-based architecture for visual attention prediction. In *2021 IEEE International Conference on Image Processing (ICIP)*. IEEE, 1464–1468.

Mohamed Amine Kerkouri, Marouane Tliba, Aladine Chetouani, and Mohamed Sayeh. 2022b. SalyPath360: Saliency and scanpath prediction framework for omnidirectional images. *Electronic Imaging* 34, 11 (2022), 168–1–168–1. https://doi.org/10.2352/EI.2022.34.11.HVEI-168

Marouane Tliba, Mohamed Amine Kerkouri, Aladine Chetouani, and Alessandro Bruno. 2022a. Self supervised scanpath prediction framework for painting images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1539–1548.

Marouane Tliba, Mohamed A. Kerkouri, Bashir Ghariba, Aladine Chetouani, Arzu Çöltekin, Mohamed Sami Shehata, and Alessandro Bruno. 2022b. SATSal: A Multi-Level Self-Attention Based Architecture for Visual Saliency Prediction. *IEEE Access* 10 (2022), 20701–20713. https://doi.org/10.1109/ACCESS.2022.3152189

David C Wong, Bin Wang, Gorkem Durak, Marouane Tliba, Mohamed Amine Kerkouri, Aladine Chetouani, Ahmet Enis Cetin, Cagdas Topel, Nicolo Gennaro, Camila Vendrami, Tugce Agirlar Trabzonlu, Amir Ali Rahsepar, Laetitia Perronne, Matthew Antalek, Onural Ozturk, Gokcan Okur, Andrew C. Gordon, Ayis Pyrros, Frank H Miller, Amir A Borhani, Hatice Savas, Eric M. Hart, Elizabeth A Krupinski, and Ulas Bagci. 2025. Shifts in Doctors' Eye Movements Between Real and AI-Generated Medical Images. In *Proceedings of the 2025 Symposium on Eye Tracking Research and Applications (ETRA '25)*. Association for Computing Machinery, New York, NY, USA, Article 106, 7 pages. https://doi.org/10.1145/3715669.3726789

Zhibo Yang, Sounak Mondal, Seoyoung Ahn, Gregory Zelinsky, Minh Hoai, and Dimitris Samaras. 2023. Predicting Human Attention using Computational Attention. *arXiv preprint arXiv:2303.09383* (2023).